Schatten–p approximation selects the strict spectral approximant for every matrix hyperplane

Candidate partial result for arXiv:2603.07498

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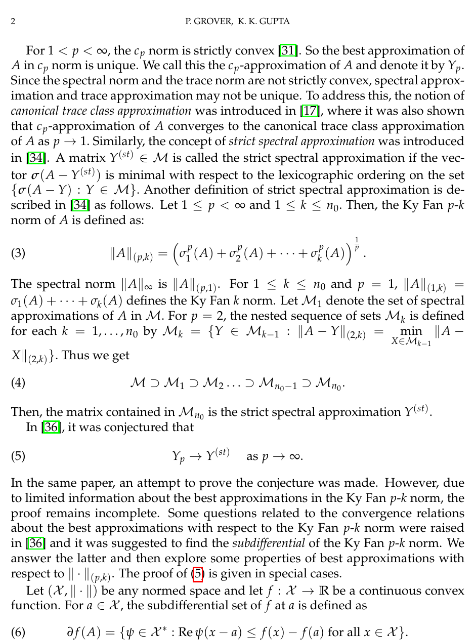
Status. `candidate_partial_likely_valid`. The source conjecture is not settled for arbitrary subspaces. We prove it for every complex codimension-one subspace, in all finite rectangular dimensions.

1 Source problem and result

For $A \in M_{m\times n}(\mathbb{C})$ and a complex linear subspace $\mathcal{M}$, let $Y_p \in \mathcal{M}$ minimize the Schatten p-norm $\|A - Y\|_p$, where $1 < p < \infty$. Grover and Gupta ask in their Conjecture (5) whether

$$Y_p \longrightarrow Y^{(\mathrm{st})} \qquad (p \to \infty),$$

where $Y^{(\mathrm{st})}$ is the strict spectral approximant: among spectral-norm minimizers, it lexicographically minimizes the complete singular-value vector of the residual.



2 P. GROVER, K. K. GUPTA

For $1 < p < \infty$, the c_p norm is strictly convex [31]. So the best approximation of A in c_p norm is unique. We call this the c_p-approximation of A and denote it by Y_p. Since the spectral norm and the trace norm are not strictly convex, spectral approximation and trace approximation may not be unique. To address this, the notion of *canonical trace class approximation* was introduced in [17], where it was also shown that c_p-approximation of A converges to the canonical trace class approximation of A as $p \to 1$. Similarly, the concept of *strict spectral approximation* was introduced in [34]. A matrix $Y^{(st)} \in \mathcal{M}$ is called the strict spectral approximation if the vector $\boldsymbol{\sigma}(A - Y^{(st)})$ is minimal with respect to the lexicographic ordering on the set $\{\boldsymbol{\sigma}(A - Y) : Y \in \mathcal{M}\}$. Another definition of strict spectral approximation is described in [34] as follows. Let $1 \le p < \infty$ and $1 \le k \le n_0$. Then, the Ky Fan p-k norm of A is defined as:

$$\|A\|_{(p,k)} = \left(\sigma_1^p(A) + \sigma_2^p(A) + \cdots + \sigma_k^p(A)\right)^{\frac{1}{p}}. \tag{3}$$

The spectral norm $\|A\|_\infty$ is $\|A\|_{(p,1)}$. For $1 \le k \le n_0$ and $p = 1$, $\|A\|_{(1,k)} = \sigma_1(A) + \cdots + \sigma_k(A)$ defines the Ky Fan k norm. Let $\mathcal{M}_1$ denote the set of spectral approximations of A in $\mathcal{M}$. For $p = 2$, the nested sequence of sets $\mathcal{M}_k$ is defined for each $k = 1, \ldots, n_0$ by $\mathcal{M}_k = \{Y \in \mathcal{M}_{k-1} : \|A - Y\|_{(2,k)} = \min_{X \in \mathcal{M}_{k-1}} \|A - X\|_{(2,k)}\}$. Thus we get

$$\mathcal{M} \supset \mathcal{M}_1 \supset \mathcal{M}_2 \ldots \supset \mathcal{M}_{n_0 - 1} \supset \mathcal{M}_{n_0}. \tag{4}$$

Then, the matrix contained in $\mathcal{M}_{n_0}$ is the strict spectral approximation $Y^{(st)}$.

In [36], it was conjectured that

$$Y_p \to Y^{(st)} \quad \text{as } p \to \infty. \tag{5}$$

In the same paper, an attempt to prove the conjecture was made. However, due to limited information about the best approximations in the Ky Fan p-k norm, the proof remains incomplete. Some questions related to the convergence relations about the best approximations with respect to the Ky Fan p-k norm were raised in [36] and it was suggested to find the *subdifferential* of the Ky Fan p-k norm. We answer the latter and then explore some properties of best approximations with respect to $\|\cdot\|_{(p,k)}$. The proof of (5) is given in special cases.

Let $(\mathcal{X}, \|\cdot\|)$ be any normed space and let $f : \mathcal{X} \to \mathbb{R}$ be a continuous convex function. For $a \in \mathcal{X}$, the subdifferential set of f at a is defined as

$$\partial f(A) = \{\psi \in \mathcal{X}^* : \operatorname{Re} \psi(x - a) \le f(x) - f(a) \text{ for all } x \in \mathcal{X}\}. \tag{6}$$

Real crop from page 2 of the source paper, including Conjecture (5).

Proof intuition. A hyperplane has a single normal matrix G. Schatten duality makes the finite-p residual the unique equality case in one Hölder inequality, so it can be written explicitly from the singular values of G. At $p = \infty$, equality in nuclear/operator duality forces the same maximal singular block in *every* spectral minimizer. The strict minimizer is then obtained by setting the unconstrained complementary block equal to zero. The explicit finite-p formula converges to precisely that matrix.

2 The hyperplane theorem

Theorem 1. *Let $G \in M_{m\times n}(\mathbb{C}) \setminus \{0\}$ and*

$$\mathcal{M} = \{X \in M_{m\times n}(\mathbb{C}) : \operatorname{tr}(G^* X) = 0\}.$$

For $A \in M_{m\times n}(\mathbb{C})$, let Y_p be its unique Schatten p-best approximant from $\mathcal{M}$, $1 < p < \infty$. Then

$$Y_p \longrightarrow Y^{(\mathrm{st})}.$$

More explicitly, set $c = \operatorname{tr}(G^* A)$. If $c = 0$, then $Y_p = Y^{(\mathrm{st})} = A$. If $c \neq 0$, take a thin singular value decomposition

$$G = U_r \operatorname{diag}(g_1, \ldots, g_r) V_r^*, \qquad g_i > 0, \quad r = \operatorname{rank} G,$$

and set $\omega = c/|c|$, $q = p/(p-1)$. Then

$$A - Y_p = \frac{\omega |c|}{\sum_{i=1}^r g_i^q}\, U_r \operatorname{diag}\left(g_1^{q-1}, \ldots, g_r^{q-1}\right) V_r^*, \tag{1}$$

$$A - Y^{(\mathrm{st})} = \frac{\omega |c|}{\sum_{i=1}^r g_i}\, U_r V_r^*. \tag{2}$$

Proof. Approximating A from $\mathcal{M}$ is equivalent to minimizing $\|R\|_p$ subject to

$$\operatorname{tr}(G^* R) = c, \tag{3}$$

where $R = A - Y$. The assertion for $c = 0$ is immediate, so suppose $c \neq 0$.

Schatten Hölder duality, with $q = p/(p-1)$, gives every feasible R the lower bound

$$|c| = |\operatorname{tr}(G^* R)| \leq \|G\|_q \|R\|_p, \qquad \|G\|_q = \Big(\sum_i g_i^q\Big)^{1/q}. \tag{4}$$

The matrix on the right side of (1) is feasible and attains equality in (4), hence is a minimizer. Since the Schatten p-norm is strictly convex for $1 < p < \infty$, the minimizer is unique. This proves (1).

It remains to identify the strict spectral residual. Put

$$\rho = \frac{|c|}{\|G\|_1} = \frac{|c|}{\sum_i g_i}.$$

Nuclear/operator duality gives $\|R\|_\infty \geq \rho$ for every feasible R, while

$$R_0 = \omega \rho\, U_r V_r^* \tag{5}$$

is feasible and has operator norm ρ. Thus ρ is the optimal spectral error.

We now determine all equality cases. Write u_i, v_i for the singular vectors in the displayed decomposition of G. If R is any spectral minimizer, then

$$\begin{aligned} |c| &= \left|\sum_{i=1}^{r} g_i\, u_i^* R v_i\right| \\ &\le \sum_{i=1}^{r} g_i |u_i^* R v_i| \le \rho \sum_{i=1}^{r} g_i = |c|. \end{aligned} \tag{6}$$

All inequalities are equalities. Because every $g_i > 0$, equality in the last inequality yields $|u_i^* R v_i| = \rho$ for every i, and equality in the triangle inequality gives their common phase:

$$u_i^* R v_i = \omega\rho. \tag{7}$$

Moreover,

$$\rho = |u_i^* R v_i| \le \|R v_i\|_2 \le \|R\|_\infty = \rho.$$

Equality in Cauchy–Schwarz implies $R v_i = \omega\rho u_i$. Applying the same argument to R^* gives $R^* u_i = \overline{\omega}\rho v_i$. Consequently, relative to the orthogonal decompositions

$$\mathbb{C}^n = \operatorname{span}\{v_i\}_{i=1}^r \oplus V^\perp, \qquad \mathbb{C}^m = \operatorname{span}\{u_i\}_{i=1}^r \oplus U^\perp,$$

every spectral-minimizing residual has block form

$$R = \begin{pmatrix} \omega\rho I_r & 0 \\ 0 & B \end{pmatrix}, \qquad \|B\|_\infty \le \rho. \tag{8}$$

Its singular-value vector therefore begins with at least r copies of ρ. Formula (5) corresponds to $B = 0$, so its vector is

$$(\underbrace{\rho, \ldots, \rho}_{r \text{ times}}, 0, \ldots, 0),$$

which is lexicographically smallest among (8). It is also the unique matrix with this vector, because zero remaining singular values force $B = 0$. Hence $R_0 = A - Y^{(\mathrm{st})}$, proving (2).

Finally, $q \downarrow 1$ as $p \to \infty$, so $g_i^{q-1} \to 1$ and $\sum_i g_i^q \to \sum_i g_i$. Formula (1) therefore converges to (2). □

3 Why this is beyond the unique spectral case

Rank-deficient G produces genuinely nonunique spectral approximations. For example, take

$$G = \operatorname{diag}(2, 1, 0), \qquad A = \operatorname{diag}(1, 1, 0), \qquad \mathcal{M} = \ker\bigl(X \mapsto \operatorname{tr}(G^* X)\bigr).$$

Here $c = 3$, $\rho = 1$, and all residuals $\operatorname{diag}(1, 1, z)$, $|z| \le 1$, are spectral minimizers. The strict residual is $\operatorname{diag}(1, 1, 0)$, whereas (1) gives

$$A - Y_p = \frac{3}{2^q + 1} \operatorname{diag}(2^{q-1}, 1, 0) \longrightarrow \operatorname{diag}(1, 1, 0).$$

4 Verification, limitations, and novelty

Computational check. The accompanying script generated 1,000 random complex hyperplanes of sizes 3×4 and 4×4, including all possible ranks. At $p = 2, 4, 10, 50, 200$, it checked feasibility, compared (1) against random feasible perturbations, and checked convergence to (2). The maximum feasibility error was $5.024 \cdot 10^{-15}$, the minimum sampled optimality slack was $5.813 \cdot 10^{-7}$, and the maximum $p = 200$ Frobenius error to the strict residual was $3.231 \cdot 10^{-2}$. These tests are not used as proof.

Limitations. The unrestricted conjecture remains open. In codimension at least two, the dual norming matrix can vary with q, so the single-normal equality argument above does not supply a uniform lexicographic selection theorem.

Novelty check. A bounded search of the run indexes, the source paper, and directly related older literature found no statement of this all-hyperplane theorem. Zię̧tak's pointwise eventual comparison theorem for a fixed competitor does not by itself control the moving minimizers Y_p. This is a preliminary novelty assessment, not a literature claim.

Human-review recommendation. Check especially the equality passage from (6) to the block form (8) and search for an existing codimension-one theorem stated in duality language. If both checks pass, the result appears ready to promote as a rigorous partial resolution.

References

[1] P. Grover and K. K. Gupta, *Properties of best approximations with respect to the Ky Fan p-k norm, and the strict spectral approximant of a matrix*, arXiv:2603.07498v2 (2026).

[2] K. Ziętak, *Strict spectral approximation of a matrix and some related problems*, Applicationes Mathematicae **44** (2017), 267–280.